



HEALTH CONNECT CLINIC — INITIAL DATA ANALYTICS ASSESSMENT

HealthConnect Clinic — Initial Data Analytics Assessment



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Table of Content

Contents

My Role Within the Broader HealthConnect Project	2
Dataset overview and Data Quality Assessment	3
Missing values — all logically explainable, not data errors:	3
Logic checks — all passed, no contradictions found:	3
Categorical values — all clean, no typos/inconsistencies:	3
One thing worth flagging as a genuine finding, not a data issue	3
Business Questions.....	5
Proposed KPIs	6
Initial Analysis Approach.....	7
Stage 1 — Data Preparation	7
Stage 2 — Exploratory Analysis by Business Question	7
Stage 3 — Synthesis and Recommendations	7
Assumptions, Limitations, Risks and Dependencies	8
Assumptions	8
Limitations	8
Risks	8
Dependencies	9
Week 4 Project Summary — Data Analytics Track	10
1. The Problem This Track Will Address.....	10
2. Resources Used	10
3. Key Observations from Week 4.....	10
4. Proposed Approach	10
5. Key Considerations	11
6. Proposed Focus for Week 5.....	11

My Role Within the Broader HealthConnect Project

As the Data Analytics track contributor, my role is to investigate patterns and drivers behind patient no-shows using the appointment dataset, and to translate those findings into clear, evidence-based business questions and KPIs. This analytical foundation is intended to directly support the wider HealthConnect project: the patterns identified here (particularly around reminder effectiveness and previous no-show history) are expected to inform the Data Science track's predictive modeling work, while insights into patient behavior and common appointment-related friction points may also inform the Generative AI track's design of a patient-facing assistant. In this sense, the Data Analytics track's Week 4 output serves as a shared analytical foundation for the other tracks' subsequent work, not an isolated deliverable.

Dataset overview and Data Quality Assessment

The HealthConnect Appointment Dataset contains 5,000 appointments × 18 columns, no duplicate rows, no duplicate appointment IDs.

Missing values — all logically explainable, not data errors:

- reminder_channel: 1,366 missing — but every single one of these corresponds exactly to reminder_sent = "No" (confirmed: no contradictions). This isn't broken data — it's expected, since there's no channel to record if no reminder was sent.
- distance_to_clinic_km: 90 missing (1.8%).
- waiting_time_minutes: 60 missing (1.2%).

Logic checks — all passed, no contradictions found:

- No patient has more previous no-shows than previous appointments (would be logically impossible) ✓
- No appointment was booked *after* it occurred ✓
- Reminder channel is never filled in when no reminder was sent ✓

Categorical values — all clean, no typos/inconsistencies:

- Gender: Female, Male, Prefer not to say (3 values).
- Age: 18-80, grouped into 6 age bands.
- Appointment type: 4 categories (Follow-up, Specialist Consultation, General Consultation, Diagnostic Test).
- Appointment outcome: **No-Show (2,423 / 48.5%)**, Attended (2,314 / 46.3%), Cancelled (263 / 5.3%).

One thing worth flagging as a genuine finding, not a data issue: nearly half of all appointments (48.5%) end in a no-show — this is a very high rate for a real clinic and confirms the business problem stated in the brief is severe, not exaggerated.

- Missing values: reminder_channel has 1,366 missing values (27.3%), but this is fully explained — every missing value corresponds exactly to reminder_sent = "No", meaning there is no data quality issue here, only an expected logical gap. distance_to_clinic_km has 90 missing values (1.8%) and waiting_time_minutes has 60 missing values (1.2%), with no clear explanatory pattern identified at this stage.
- Duplicates: No duplicate rows and no duplicate appointment_id values were found across all 5,000 records.
- Logical consistency checks: No patient records show more previous no-shows than previous appointments (which would be logically impossible), and no appointment shows a booking date occurring after its appointment date — both checks passed with zero violations.
- Categorical value consistency: All categorical fields (gender, age_group, appointment_type, appointment_day, appointment_time, reminder_sent, appointment_outcome) contain clean, consistent values with no typos or unexpected entries.
- Outcome distribution: appointment_outcome splits into No-Show (2,423 records, 48.5%), Attended (2,314 records, 46.3%), and Cancelled (263 records, 5.3%) — notably, the no-show rate is unusually high compared to typical real-world clinic benchmarks, confirming the severity of the business problem as described in the project brief.

Business Questions

1. **What is the overall no-show rate, and how does it compare to attended and cancelled appointments?** (Baseline — we already know it's 48.5%, a striking number worth framing as the headline problem.)
2. **Does distance to the clinic affect the likelihood of a no-show?** (Tests whether access/travel burden is a driver.)
3. **Does sending a reminder — and which channel (SMS, WhatsApp, Email) — reduce no-shows?** (Directly tests whether the clinic's existing reminder system is working.)
4. **Does a patient's history of previous no-shows predict future no-shows?** (Tests whether past behavior is a strong signal — this is often the single strongest predictor in real no-show data.)
5. **Does the time between booking and the appointment (booking_lead_days) affect attendance?** (Tests whether long waits between booking and the actual appointment cause people to forget or lose interest.)
6. **Do certain appointment types, days of the week, or times of day have higher no-show rates?** (Operational — helps the clinic think about scheduling patterns.)
7. **Does age or age group relate to no-show likelihood?** (Demographic angle — younger vs older patients may behave differently.)

Proposed KPIs (3-5 required, with justification — no calculation needed yet, per this week's scope)

KPI	Business Question It Supports	Why It Matters
Overall No-Show Rate (% of appointments with outcome = No-Show)	Q1	The core metric the whole project exists to improve — the single most important number for the clinic to track over time.
No-Show Rate by Reminder Status/Channel	Q3	Tests whether the clinic's current reminder system is effective, and which channel works best — directly actionable (the clinic can change reminder policy based on this).
No-Show Rate by Previous No-Show History	Q4	If previous no-shows strongly predict future ones, this becomes the clinic's most powerful early-warning signal — highly relevant for both analytics <i>and</i> feeds directly into the Data Science track's prediction model.
Average Distance to Clinic (No-Show vs Attended)	Q2	Tests whether a practical, addressable barrier (travel distance) is driving no-shows — if true, this could justify solutions like telehealth options for distant patients.
No-Show Rate by Booking Lead Time	Q5	Identifies whether short-notice or long-notice bookings behave differently, informing scheduling policy (e.g., should the clinic avoid booking appointments too far in advance?).

Initial Analysis Approach

The analysis will proceed in three stages across the coming weeks, building directly on this week's data quality assessment and business questions.

Stage 1 — Data Preparation

Address the missing values identified in `distance_to_clinic_km` (1.8%) and `waiting_time_minutes` (1.2%) using median imputation, given the small proportion affected and the absence of any clear pattern explaining their absence. `reminder_channel`'s missing values will be retained as-is, since they are logically explained by `reminder_sent = "No"` and do not represent a data quality issue. Categorical fields (`appointment_type`, `appointment_day`, `appointment_time`, `gender`, `age_group`) require no cleaning, having already been confirmed consistent with no typos or invalid values.

Stage 2 — Exploratory Analysis by Business Question

Each proposed KPI will be calculated and broken down by its relevant dimension, following the same pivot-table and visualization approach used in prior AnalystLab projects (Weeks 1-3):

- No-show rate will be calculated overall, then segmented by reminder status/channel, previous no-show history, distance bands, booking lead-time bands, appointment type, day of week, and age group.
- Particular attention will be paid to `previous_no_shows`, since patient-level historical behavior is likely to be the strongest single predictor of future no-shows, based on patterns typically observed in appointment attendance data.

Stage 3 — Synthesis and Recommendations

Findings will be synthesized into a prioritized set of business insights and recommendations (following the same evidence-based, impact-ranked format used in the Superstore project), focused specifically on actionable interventions — for example, targeted reminder strategies for high-risk patients, or scheduling adjustments for consistently high-no-show time slots.

This approach intentionally mirrors the analytical structure used in Weeks 1-3, applying the same rigor (data quality first, business-question-driven KPIs, evidence-based prioritized recommendations) to a healthcare context.

Assumptions, Limitations, Risks and Dependencies

Assumptions

- The dataset is assumed to be a representative, complete sample of HealthConnect's appointment activity, with no systematic exclusion of certain patient types or appointment categories.
- `appointment_outcome` is assumed to be recorded accurately and consistently at the time of each appointment, with "No-Show" reliably distinguished from "Cancelled" (i.e., a cancellation is not miscoded as a no-show, or vice versa).
- The dataset is fictional and synthetic, as stated in the project brief — findings and patterns are illustrative for this exercise and would need validation against real clinic data before informing actual policy decisions.

Limitations

- `distance_to_clinic_km` (1.8% missing) and `waiting_time_minutes` (1.2% missing) contain a small number of missing values with no clear explanatory pattern identified at this stage; any analysis using these fields should account for this gap.
- The dataset does not include information on *why* a patient did not attend (e.g., illness, transportation failure, forgetfulness) — only the outcome itself, limiting the depth of root-cause analysis possible from this data alone.
- The dataset appears to cover a limited time window (appointments spanning 2025-2026 based on the sampled dates); seasonal or long-term trend analysis may be constrained by the available date range.
- As a single-clinic dataset, findings may not generalize to other clinic locations or broader healthcare contexts without further validation.

Risks

- If `previous_no_shows` proves to be a strong predictor (as anticipated), there is a risk of over-relying on patient history in a way that could feel punitive or create a self-fulfilling pattern (e.g., deprioritizing patients already flagged as high-risk) — this is a consideration relevant to the Generative AI and Data Science tracks' work as the project progresses, not just Analytics.

- Recommendations based on this dataset must be clearly framed as *illustrative* given its fictional/synthetic nature, to avoid the risk of the project being mistaken for guidance based on real patient data.

Dependencies

- This track's analysis feeds directly into the Data Science track's no-show prediction model (shared KPIs like previous no-show history and reminder effectiveness are likely candidate features for their model) and the Generative AI track's assistant design (understanding common patient questions/behaviors around appointments).
- Deeper analysis in subsequent weeks depends on the missing-value handling decisions made in Stage 1 of the proposed approach being finalized and agreed upon, since they affect downstream KPI calculations.

Week 4 Project Summary — Data Analytics Track

1. The Problem This Track Will Address

As the Data Analytics track for the HealthConnect project, this work focuses on understanding the patterns and drivers behind patient appointment no-shows — currently affecting 48.5% of all appointments — using the available appointment dataset to surface actionable, evidence-based insights that support the clinic's central goal of reducing missed appointments.

2. Resources Used

- HealthConnect_Appointment_Data.csv — 5,000 appointment records across 18 variables
- HealthConnect_Data_Dictionary.xlsx — used to understand variable definitions and structure before analysis began

3. Key Observations from Week 4

- The dataset is well-structured and largely clean: no duplicate records, no logical contradictions (e.g., no patient shows more previous no-shows than previous appointments), and only minor missing values (under 2%) in two numeric fields (distance_to_clinic_km, waiting_time_minutes).
- The no-show rate (48.5%) is unusually high compared to typical real-world clinic benchmarks, confirming the business problem as stated is severe rather than exaggerated.
- The dataset contains several strong candidate variables for explaining no-shows — particularly reminder status/channel, previous no-show history, distance to clinic, and booking lead time — each of which maps directly to a specific, testable business question.

4. Proposed Approach

A three-stage approach is proposed for subsequent weeks: (1) minor data preparation to address the two fields with missing values, (2) exploratory analysis breaking down no-show rate by each of the 5 proposed KPI dimensions, and (3) synthesis into prioritized, evidence-based recommendations — mirroring the analytical structure applied in prior AnalystLab projects.

5. Key Considerations

The dataset's synthetic/fictional nature means findings are illustrative rather than directly actionable for a real clinic. The absence of "reason for no-show" data limits root-cause depth. This track's outputs (particularly around previous no-show history and reminder effectiveness) are expected to directly inform the Data Science track's predictive modeling work in later weeks.

6. Proposed Focus for Week 5

Week 5 will focus on executing Stage 1 (data preparation) and beginning Stage 2 (exploratory analysis), starting with the no-show rate breakdown by previous no-show history and reminder channel, since these are expected to be the most impactful and actionable findings for the clinic.